

Task Questions

1. Question 1

When we export a value from one stack to another, we first define the value in the CloudFormation template of the source stack, giving it a name using the Export attribute. For instance, if we want to export a VPC ID, we include it in the Outputs section of the template and name it. Then, when you import the exported value using the Fn::ImportValue function in the CloudFormation template of the target stack, we specify the name we gave it when exporting.

2. Question 2

I should set up a deletion policy in the CloudFormation template if I want to keep an EBS disk after the template is deleted. This policy outlines the steps that AWS CloudFormation must take in order to destroy the stack. The deletion policy can be configured to "Snapshot," for instance, instructing CloudFormation to take a snapshot of the EBS disk prior to erasing it. I add the DeletionPolicy attribute in the CloudFormation template and set it to "Snapshot" for the appropriate resource. This guarantees that despite the stack being erased, a snapshot of the EBS disk is kept, enabling me to keep its contents.

3. Question 3

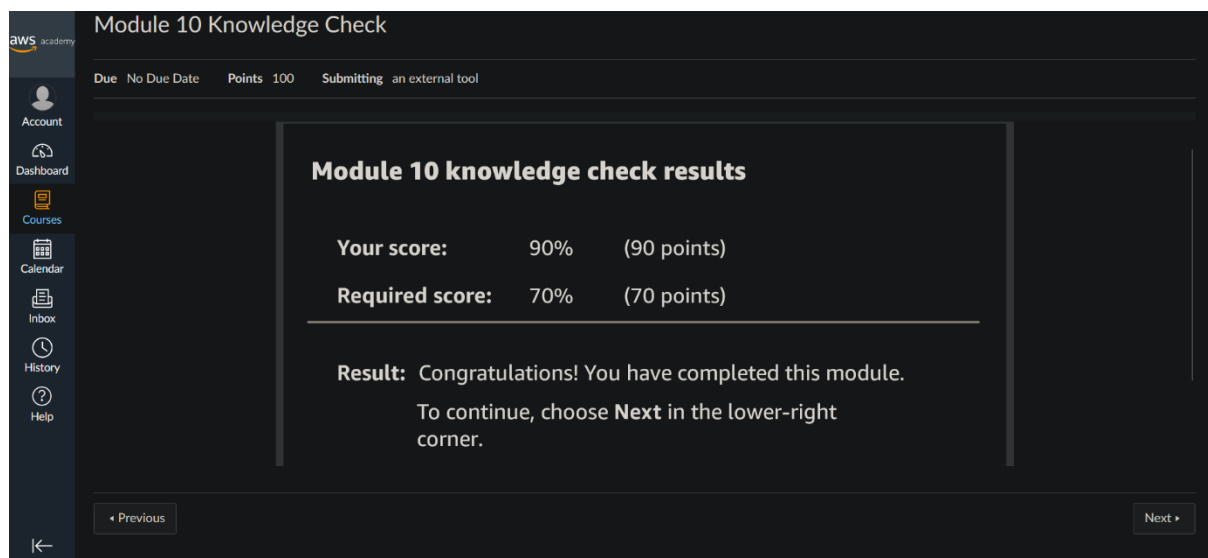
The Ref function is one example. It gives back the resource or parameter value that was supplied. For example, I may use !Ref MyEC2Instance in my CloudFormation template to refer to the name of an EC2 instance. I can refer to the resource elsewhere in the template because this function retrieves its logical name.

The function Fn::GetAtt is a further example. It returns the value of an attribute from a resource in the template. For instance, I can use to obtain an EC2 instance's public IP address. Acquire MyEC2Instance.PublicIp. I may access particular resource properties, like IP addresses, ARNs, or availability zones, within the template with the help of this function.

Module 10 Knowledge check

This knowledge check was comprised of the most important questions throughout the module 10. The questions asked was kind of a summary to the whole module 10. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about Automating your Architecture. First question itself was about automating and provisioning resources. Then there were questions about AWS CloudFront and CloudFront designer. Then there were questions about how to use automation efficiently and securely. Then there was a question about AWS OpsWork which provides managed chef instances, automates server configurations, deployment, management and also provides managed puppet instances. Finally there was a question about elastic bean stalk and real life scenarios.



The screenshot shows the AWS Academy interface for the 'Module 10 Knowledge Check'. The top navigation bar includes 'Account', 'Dashboard', 'Courses', 'Calendar', 'Inbox', 'History', and 'Help'. The main content area displays the following information:

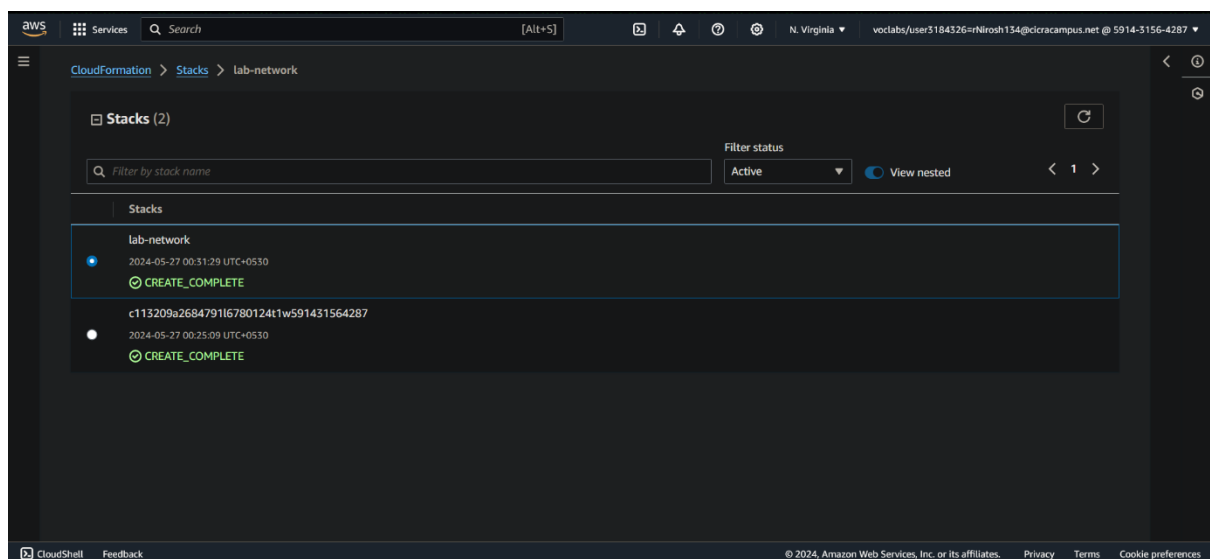
- Module 10 Knowledge Check**
- Due:** No Due Date
- Points:** 100
- Submitting:** an external tool
- Module 10 knowledge check results**
- Your score:** 90% (90 points)
- Required score:** 70% (70 points)
- Result:** Congratulations! You have completed this module. To continue, choose **Next** in the lower-right corner.

At the bottom, there are navigation buttons: 'Previous' and 'Next'.

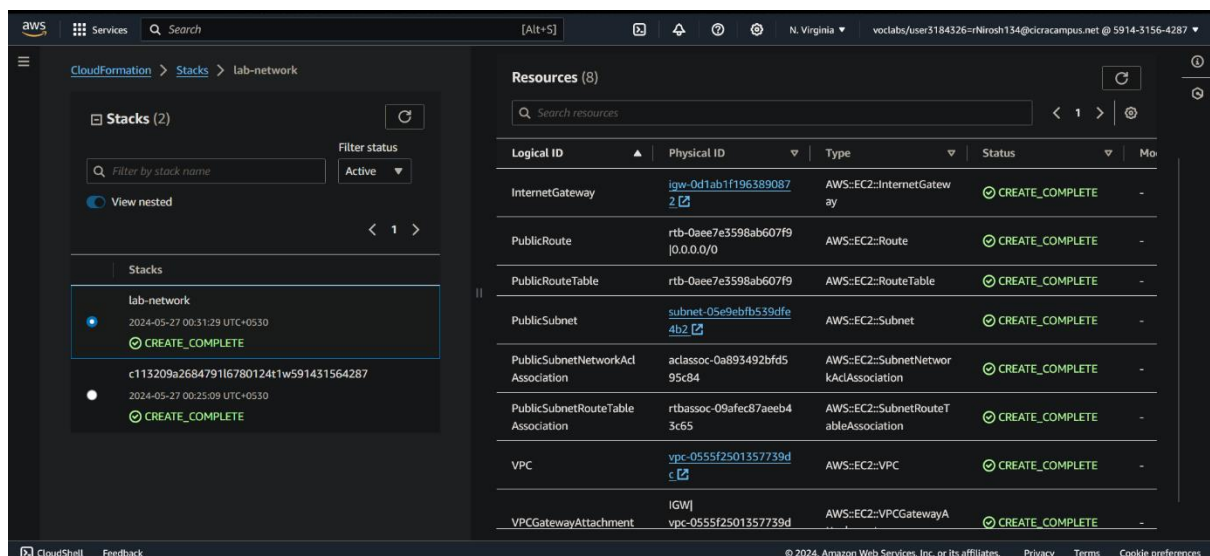
Module 10 Guided Lab – Automating Infrastructure Deployment with AWS CloudFormation

Task 1: Deploying a networking layer

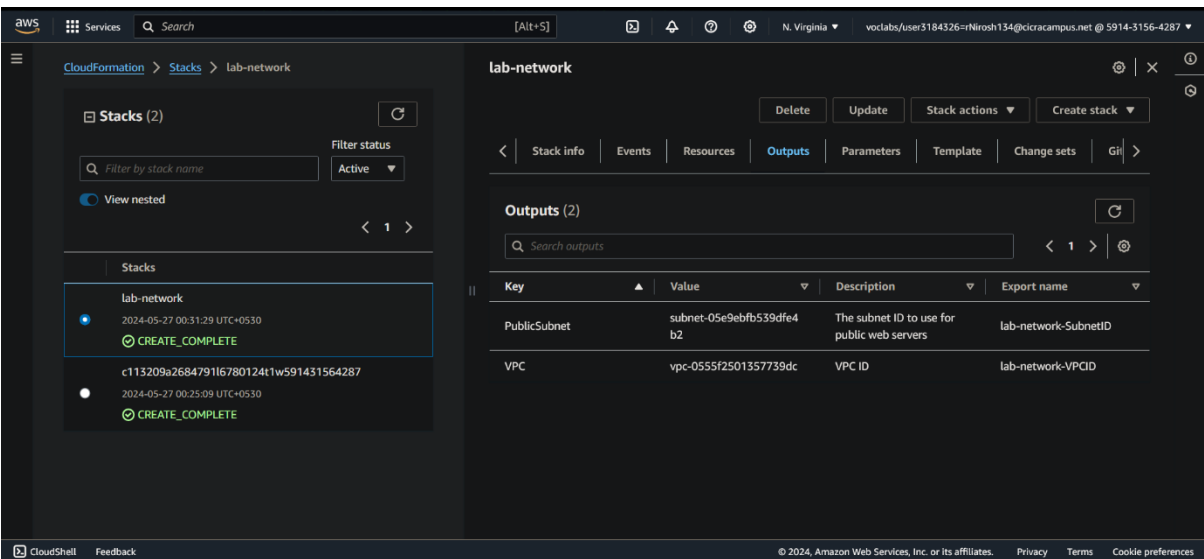
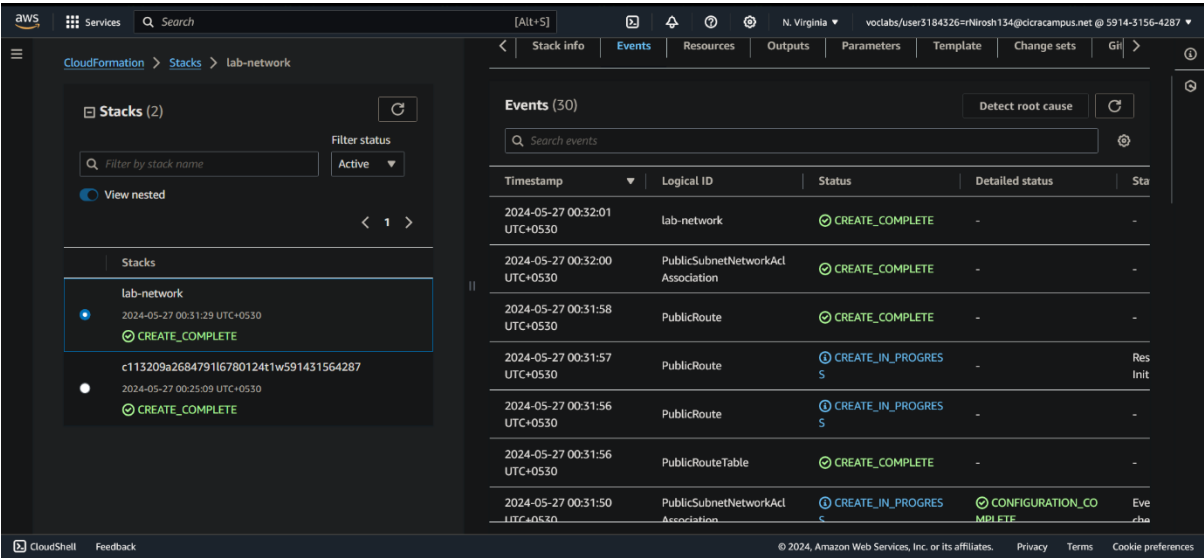
I'm setting up a network layer in AWS using a CloudFormation template, which helps me reuse the setup across different environments. First, I download the lab-network.yaml template file and open it in a text editor to see the AWS resources it defines.



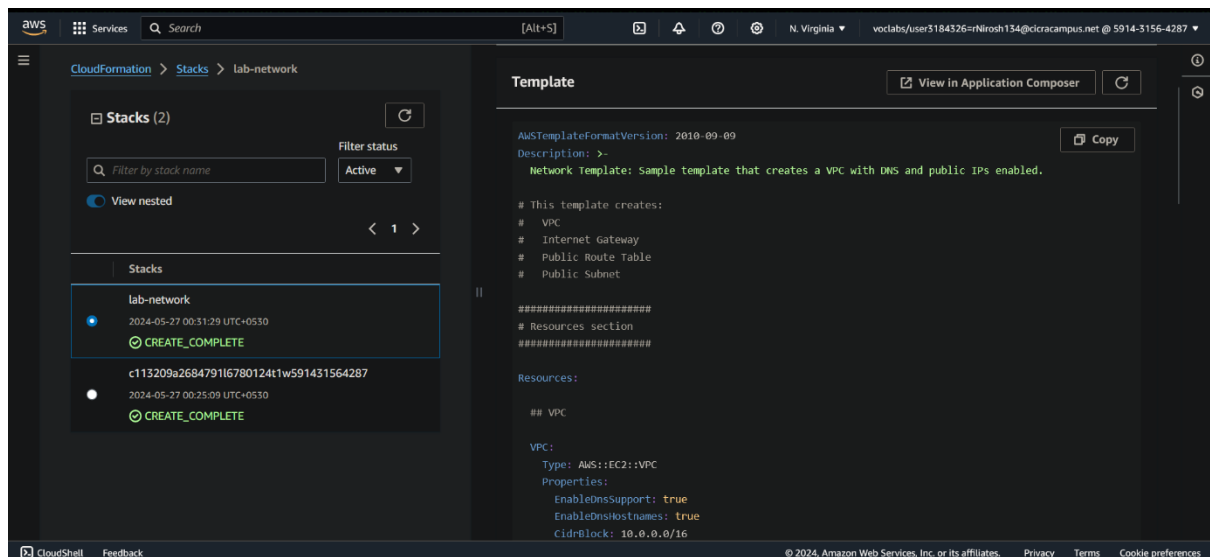
In the AWS Management Console, I navigate to CloudFormation, click Create stack button and upload the template file.



I name the stack lab-network, add a tag with Key application and Value inventory, and create the stack. I then monitor the stack creation process, waiting for the status to change to CREATE_COMPLETE.

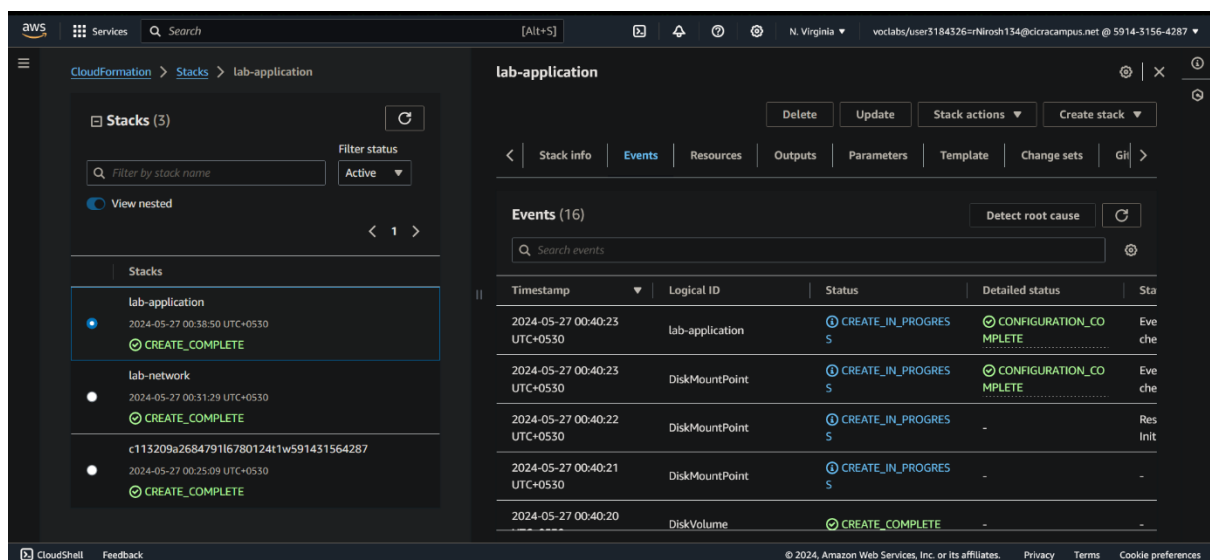


Once complete, I check the Resources tab to see the created resources, the Events tab for the creation log, and the Outputs tab for important resource IDs like the public subnet and VPC. Finally, I review the template used for the stack in the Template tab.

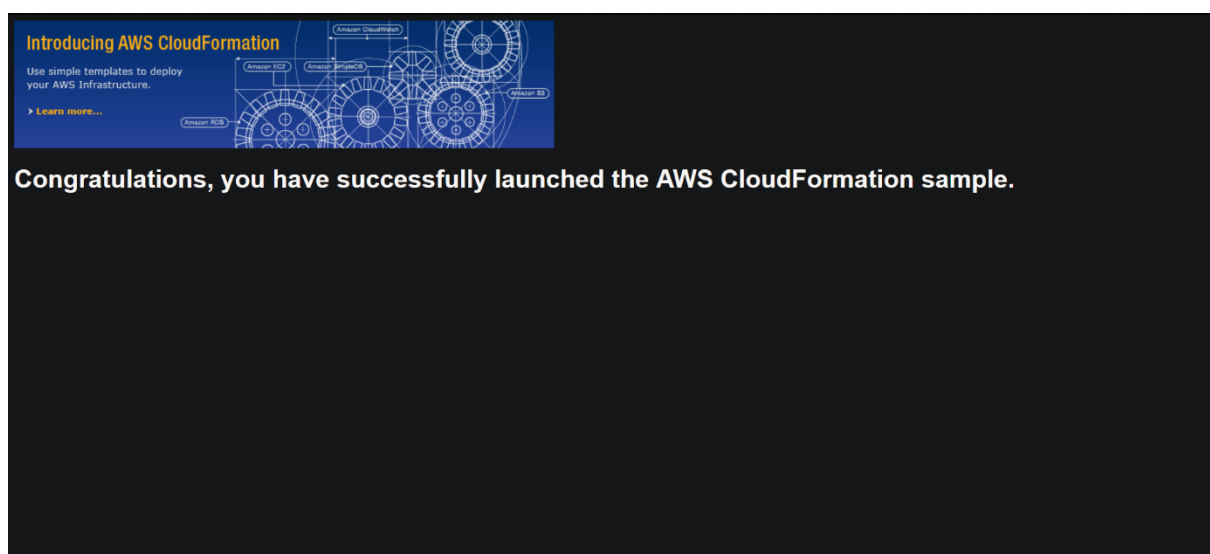


Task 2: Deploying an application layer

I'm setting up an application layer in AWS, which includes an EC2 instance and a security group. The CloudFormation template uses the VPC and subnet IDs from my previously created network stack. First, I download the lab-application.yaml template file and open it if I want to see the resource definitions. In the AWS Management Console, I go to CloudFormation, select Create stack with new resources, and upload the template file. I name the stack lab-application and specify lab-network as the network stack name. I add a tag with Key application and Value inventory, review the details, and create the stack. I monitor the stack creation in the Events and Resources tabs and wait for the status to change to CREATE_COMPLETE.

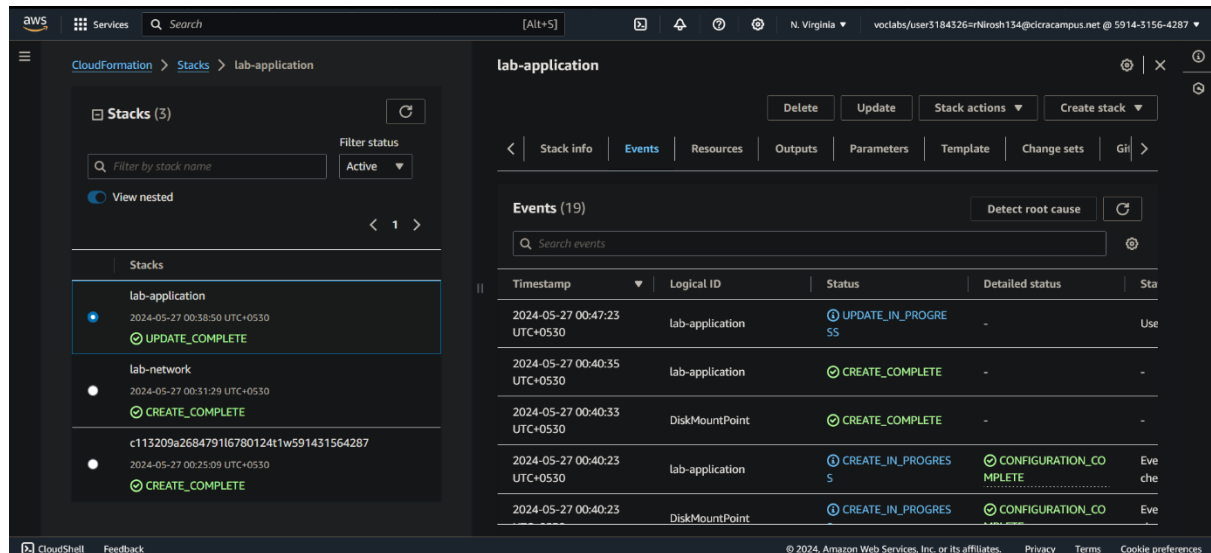


Once done, I go to the Outputs tab, copy the provided URL, and open it in a browser to see my application running. The template ensures the EC2 instance and security group are correctly placed in the VPC and subnet created by the network stack.

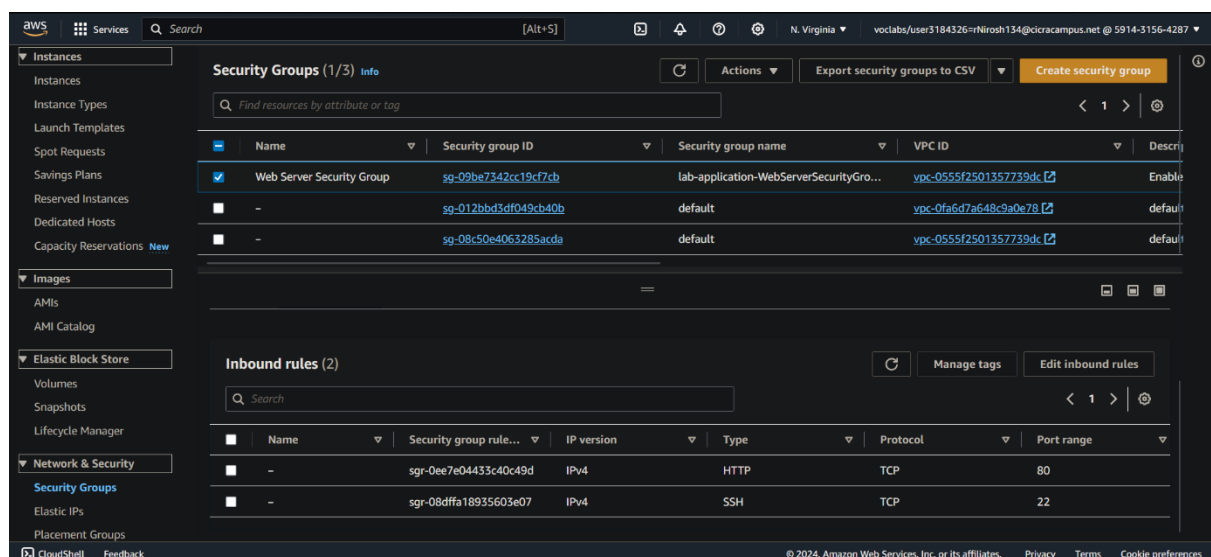


Task 3: Updating a Stack

I'm updating my application stack to change the security group settings. First, I check the current security group settings in the EC2 console, which currently allows only HTTP traffic. Then, I download a new template (lab-application2.yaml) that adds a rule for SSH traffic on port 22. In the CloudFormation console, I select the lab-application stack, choose Update, and upload the new template.

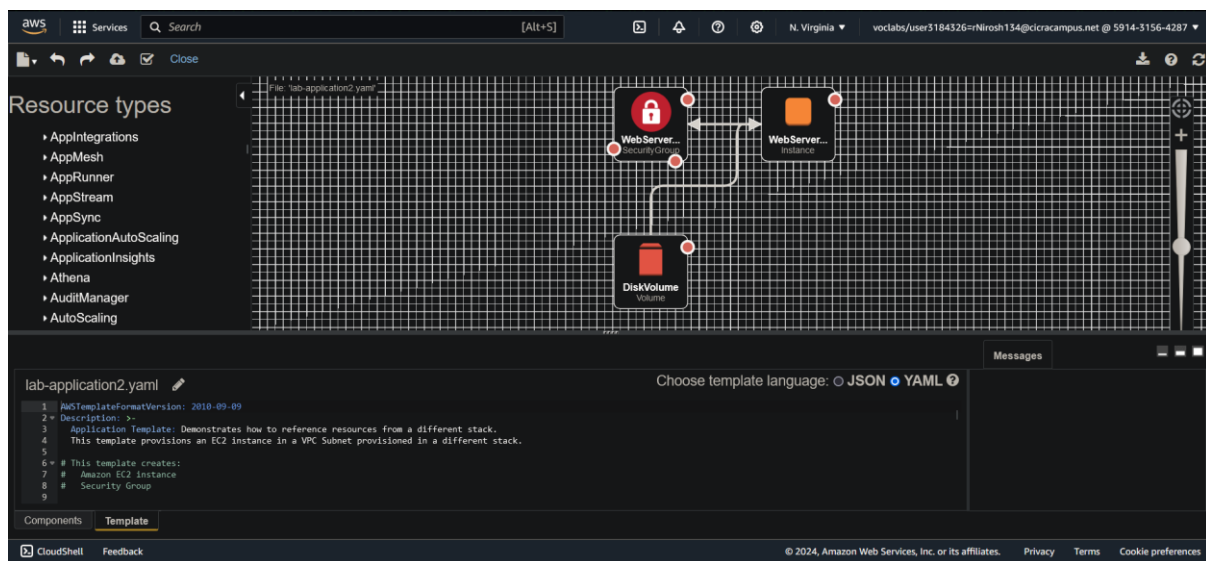


I review the changes and update the stack. Once the update is complete, I verify the new rule in the EC2 console, confirming that SSH traffic is now allowed. This process shows how I can update infrastructure in a repeatable and documented way using CloudFormation templates.



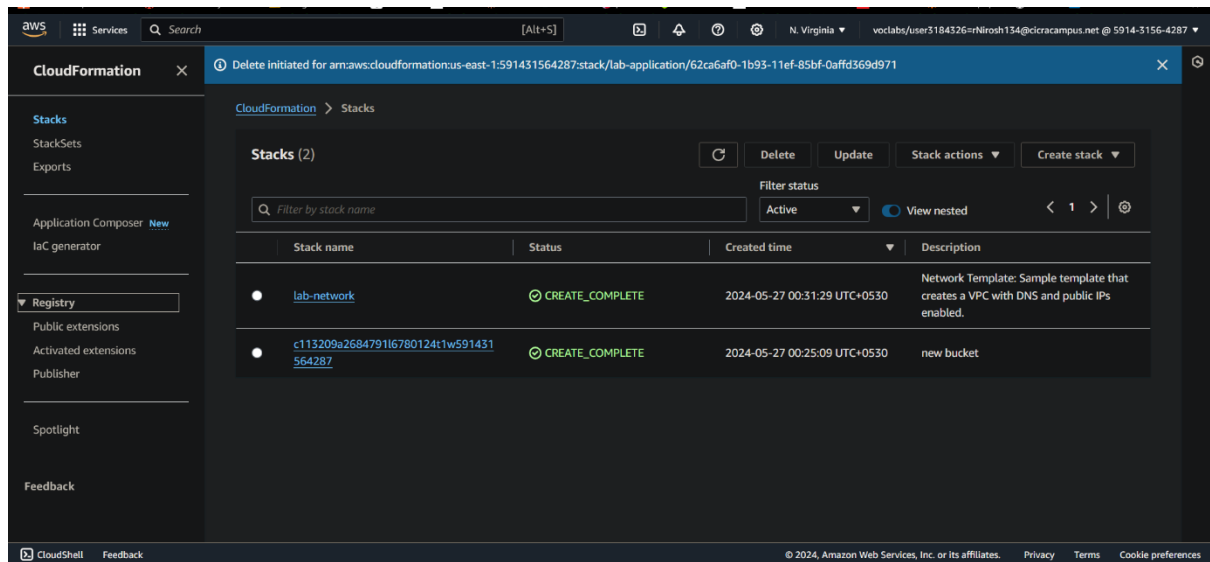
Task 4: Exploring templates with AWS CloudFormation Designer

I'm exploring AWS CloudFormation Designer, which lets me create, view, and modify CloudFormation templates graphically. First, I open CloudFormation in the AWS Management Console and select "Designer" from the left menu. I then load the lab-application2.yaml template, which displays a visual representation of the resources and their relationships. I click on resources to see their definitions, drag new resources into the design area to add them to the template, and connect resources to define relationships. I also open the lab-network.yaml template to explore its resources. This tool helps me quickly understand and modify templates.

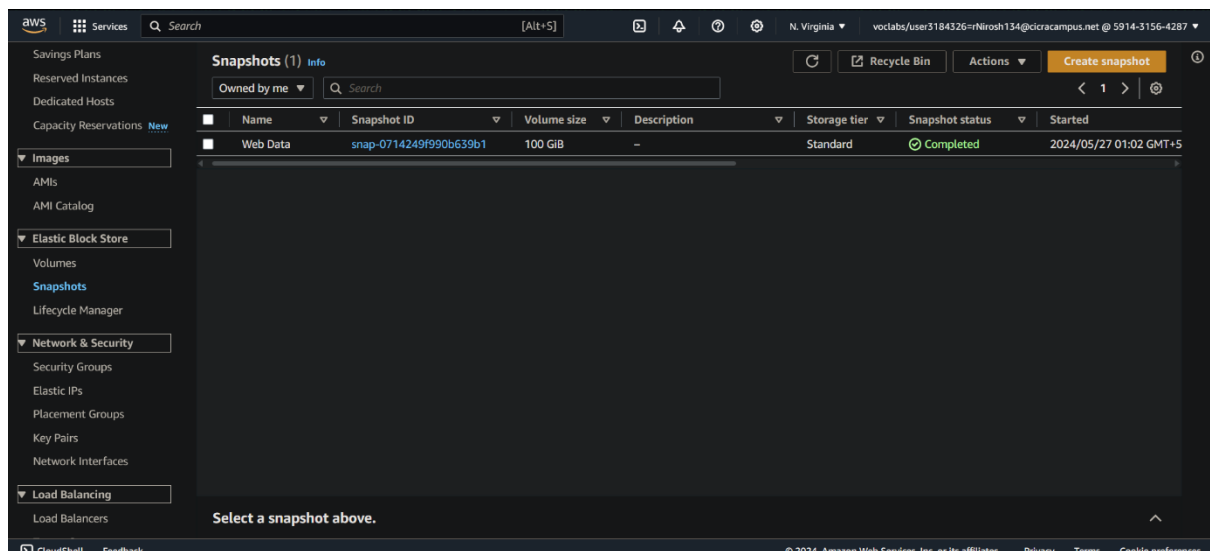


Task 5: Deleting the stack

I'm going to delete the lab-application stack in AWS CloudFormation and see how it handles creating a snapshot of the Amazon EBS volume before deletion. First, I close the Designer page and return to the main CloudFormation console. I select the lab-application stack, click Delete, and confirm by choosing Delete stack. I monitor the deletion process in the Events tab and refresh the screen occasionally.



After the stack is deleted, I switch to the EC2 service and check Snapshots in the left menu. I should see a recent snapshot, showing that the EBS volume was backed up before deletion. This process removes the application stack but leaves the network stack intact, allowing different teams to manage their own stacks.



SIT233 – Cloud Computing
Module 10 Guided Lab - Automating Infrastructure
Deployment with
AWS CloudFormation

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Grade:

My grade is 20/20

aws academy

ACAv2EN... > Assignments

> Module 10 Guided Lab - Automating Infrastructure Deployment with AWS CloudFormation

Module 10 Guided Lab - Automating Infrastructure Deployment with AWS CloudFormation

Due No Due Date Points 100 Submitting an external tool

Submit Details AWS Start Lab End Lab 2:20 Instructions Grades Actions

Files README Terminal Source

EN_US

You will now verify that a snapshot of the EBS volume was created before the EBS volume was deleted.

43. From the **Services** menu, choose **EC2**.

44. In the left navigation pane, choose **Snapshots**.

You should see a snapshot with a **Started** time in the last few minutes.

Submitting your work

45. At the top of these instructions, choose **Submit** to record your

```
bash
Testing report - The lab-network stack was successfully cr
Testing report - The lab-application stack was successfull
Testing report - The lab-application stack was successfull
Testing report - The lab-application stack was successfull
gradeFile - /mnt/vocwork3/grader/eee_g_2362885/asn2684790_
OSTI
reportFile - /mnt/vocwork3/grader/eee_g_2362885/asn2684790_
Y275
/mnt/vocwork3/grader/eee_g_2362885/asn2684790_13/asn268479
/usr/lib/python3.7/site-packages/boto3/compat.py:82: Pytho
r support Python 3.7 starting December 12, 2023. To continue r
security updates please upgrade to Python 3.8 or later. More
ws.amazon.com/blogs/developer/python-support-policy-updates-fo
arnings.warn(warning, PythonDeprecationWarning)
len 4
Present working directory = /mnt/vocwork3/grader/eee_g_236
1/work
Default region: us-east-1
Back in submit.sh...
end
eee_W_3866282@runweb125275:~$
eee_W_3866282@runweb125275:~$
```

Total score 20/20

[Task 1] Deploy Networking Stack 5/5

[Task 2] Deploy Application Stack 5/5

[Task 3] Update Application Stack 5/5

[Task 5] Delete Application Stack 5/5